

KRASNOYARSK INTL, Russian Federation

WMO: 284935

Lat: **56.1795N**Lon: **92.4715E**Elev: **287**StdP: **97.92**Time Zone: **7.00 (E07)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.8	-35.8	-41.8	0.1	-38.5	-38.9	0.1	-35.6	10.1	-9.3	9.2	-9.9	0.2	110	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	29.8	19.2	27.9	18.2	26.0	17.2	20.5	26.9	19.5	25.5	18.4	24.3	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.5	22.8	17.1	12.7	22.0	16.1	11.8	21.1	60.1	26.4	56.4	25.5	53.2	24.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.1	7.2	DB	-38.9	33.0	4.5	1.7	-42.2	34.2	-44.9	35.2	-47.4	36.2	-50.7	37.5	
			WB	-39.0	22.3	4.5	1.6	-42.3	23.4	-44.9	24.4	-47.4	25.3	-50.6	26.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-18.1	-14.8	-5.5	2.4	9.2	16.4	18.1	15.5	9.0	1.2	-8.5	-15.1
	DBStd	14.06	9.64	8.53	6.33	5.18	5.29	3.90	2.87	3.58	4.44	5.63	8.12	9.20
	HDD10.0	4043	871	694	480	233	81	4	0	3	72	276	554	777
	HDD18.3	6430	1129	927	738	478	286	80	39	101	281	531	804	1035
	CDD10.0	727	0	0	0	4	56	196	252	174	41	3	0	0
	CDD18.3	71	0	0	0	0	3	22	32	14	1	0	0	0
	CDH23.3	1117	0	0	0	1	86	387	413	204	26	0	0	0
	CDH26.7	286	0	0	0	0	21	100	108	52	4	0	0	0
(14) Wind	WSAvg	2.8	2.2	2.6	3.4	3.7	3.5	2.4	2.1	2.1	2.6	3.0	3.0	2.6
(15) Precipitation	PrecAvg	505	23	21	20	29	47	58	53	75	53	46	40	34
	PrecMax	632	46	39	47	57	93	96	115	139	101	84	60	60
	PrecMin	386	1	6	1	7	12	18	8	33	15	14	26	14
	PrecStd	70	12	9	13	14	27	23	25	29	23	19	9	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	4.0	11.0	21.1	29.2	31.8	32.0	30.9	26.9	18.0	10.8	3.1
		MCWB	-0.6	0.6	4.4	10.0	17.8	19.5	20.2	20.4	16.9	10.2	6.4	-0.3
	2%	DB	-1.0	1.2	7.9	17.1	25.8	29.1	29.8	28.0	22.8	14.9	6.9	0.8
		MCWB	-2.9	-1.7	2.9	8.6	14.8	18.4	19.5	19.2	14.9	9.1	3.6	-1.5
	5%	DB	-2.8	-0.2	5.8	14.2	23.0	27.2	27.8	25.8	19.8	12.0	4.1	-1.0
		MCWB	-4.5	-2.5	1.7	7.3	13.1	17.5	18.6	17.9	13.4	7.6	1.7	-2.7
	10%	DB	-4.4	-2.6	3.2	11.9	20.1	25.8	25.8	23.1	16.9	9.2	2.0	-2.9
		MCWB	-5.8	-4.4	0.3	5.9	11.5	16.6	17.8	16.6	11.8	6.2	0.3	-4.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	0.7	5.5	11.3	18.7	21.2	22.5	21.7	17.6	11.7	6.6	0.3
		MCDB	0.8	3.0	9.8	18.5	27.7	28.4	28.6	28.8	25.6	16.2	9.8	2.2
	2%	WB	-2.9	-1.2	3.3	9.6	15.9	19.8	20.7	19.9	15.4	9.7	3.7	-1.4
		MCDB	-1.2	1.0	7.0	16.0	23.2	26.8	26.6	25.6	21.3	14.0	6.4	0.4
	5%	WB	-4.4	-2.7	1.8	7.9	14.0	18.6	19.7	18.4	14.0	8.1	1.7	-2.8
		MCDB	-2.8	-0.7	5.1	13.7	20.8	25.6	25.2	24.2	18.7	11.4	3.7	-1.3
	10%	WB	-6.1	-4.4	0.4	6.0	12.3	17.5	18.7	17.3	12.5	6.3	0.2	-4.4
		MCDB	-4.6	-2.5	3.1	11.1	19.1	23.7	23.9	22.2	16.3	9.2	1.9	-2.8
(35) Mean Daily Temperature Range	MDBR		9.4	10.4	11.0	11.3	13.2	13.6	12.2	12.1	10.9	8.8	8.4	8.8
	5% DB	MCDBR	9.5	10.8	13.1	16.7	18.9	17.4	16.2	16.6	16.6	13.2	8.8	8.7
		MCWBR	8.3	9.0	9.2	9.8	10.1	7.9	7.4	8.5	9.4	8.4	6.4	7.6
	5% WB	MCDBR	9.3	10.2	12.2	15.7	16.4	14.9	13.5	14.8	14.4	12.2	8.4	8.8
		MCWBR	8.3	8.7	9.0	9.8	9.6	7.5	7.2	8.0	8.6	8.1	6.4	7.9
(40) Clear-Sky Solar Irradiance	taub		0.283	0.316	0.362	0.392	0.411	0.407	0.430	0.410	0.369	0.350	0.317	0.283
	taud		2.020	2.002	1.957	2.120	2.200	2.305	2.253	2.276	2.342	2.236	2.057	1.964
	Ebn at Noon		575	710	770	817	827	836	805	796	779	678	520	453
	Edh at Noon		65	100	135	134	132	120	125	114	93	79	63	54
(44) All-Sky Solar Radiation	RadAvg		0.78	1.77	3.12	4.30	4.92	5.94	5.31	4.25	2.71	1.43	0.71	0.46
	RadStd		0.09	0.12	0.23	0.28	0.52	0.50	0.35	0.37	0.36	0.11	0.10	0.06

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page